

Rayan Ahsan

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EDUCATION

University of Toronto

Toronto, ON

B.A.Sc. in Computer Engineering, Minor in Artificial Intelligence and Business Engineering Sep. 2023 - Apr. 2028

- **Relevant Coursework:** Natural Language Processing, Neural Networks (CNN, RNN, Transformers), Machine Learning, Data Structures and Algorithms, Object-Oriented Programming, Probability and Statistics, Data Analysis, Agile Practices

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript/TypeScript

Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, Flask, Django, React, Next.js

Developer Tools: Git, Docker, Linux, AWS, Google Cloud Platform, Azure, MLflow

Libraries: NumPy, pandas, Matplotlib, OpenCV, SHAP, Hugging Face

EXPERIENCE

ML Developer, Project: Aircraft Anomaly Detection

Sep. 2024 – April 2025

UofT Machine Intelligence Student Team

Toronto, ON

- Designed **STFT-based signal feature-extraction pipeline** and **spectral anomaly detection** for **50K+ aircraft sensor signals**, improving anomaly detection accuracy by **27%**.
- Trained and fine-tuned **ConvVAE** on real + synthetic data, reducing false positives by **35%**.
- Deployed anomaly scoring for predictive maintenance, extending component life by **20+ cycles**.

PROJECTS

Fine-Tuned LLM Cardiology Bot | *Python, Hugging Face Transformers, JavaScript, Git*

- Fine-tuned **Meta's LLaMA-2-7B** for cardiology, reaching **0.83 answer relevancy** across **100 evaluation prompts**.
- Leveraged **PEFT** and **QLoRA** for efficient fine-tuning, reducing GPU usage while maintaining performance.
- Built a custom **JavaScript UI** to interact with the model, enabling intuitive question input and response display.
- Implemented **in-memory caching** to reduce response latency by **49%**, improving user experience.

Skin Lesion Classification using Deep Learning | *Python, PyTorch, ResNet18, Grad-CAM, Git*

- Conducted EDA and data processing on **HAM10000 dataset (10K+ images)** to enhance training dataset quality.
- Fine-tuned **ResNet18** hyperparameters with **Bayesian search**, boosting test accuracy to **79%**.
- Applied **Grad-CAM** to generate heatmaps to **visualize model decision regions**.
- Built and deployed an interactive **Streamlit app** for end-to-end image classification and results exploration.

IntelliCane – MakeUoft Hackathon Winner | *Python, YOLOv8, Arduino, Raspberry Pi*

- Developed a smart cane to assist the visually impaired, integrating **Arduino UNO, Raspberry Pi, and AI models** for real-time navigation.
- Achieved **81% object detection accuracy** using **YOLOv8** on Raspberry Pi for real-time detection.
- Programmed ultrasonic sensors on Arduino to detect obstacles up to **4m**, triggering audio alerts via buzzers.

Attention Is All You Need Implementation | *Python, PyTorch, NumPy*

- Implemented the Transformer architecture (Encoder, Multi-head attention, etc) from scratch in PyTorch.
- Applied core concepts such as **scaled dot-product attention** and **positional encoding**.
- Achieved **BLEU score of 25.7** on **WMT English–German dataset**, outperforming baseline seq2seq models.

CERTIFICATES

Microsoft Certified: Azure AI Fundamentals

Aug. 2024

MATLAB Certified

Nov. 2023

IBM Applied AI Professional Certificate

Jan. 2022